CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 6: Triangles

Topic-wise short notes with solved examples

Similar figures have the same shape but not necessarily the same size. This chapter covers similar triangles, the Basic Proportionality Theorem and the three tests for similarity.

1 Similar figures

- Two figures are **similar** if they have the same shape. They need not have the same size.
- Two figures are **congruent** if they have the same shape and the same size. All congruent figures are similar, but similar figures need not be congruent.
- All circles are similar. All squares are similar. All equilateral triangles are similar.
- Two **polygons** with the same number of sides are similar if (i) their corresponding angles are equal **and** (ii) their corresponding sides are in the same ratio. Both conditions are needed.

REMEMBER

A square and a rectangle have equal angles but are not similar, because their sides are not in the same ratio. A square and a rhombus have sides in the same ratio but are not similar, because their angles are not equal.

2 Similar triangles

- $\triangle ABC \sim \triangle DEF$ means $\angle A = \angle D, \angle B = \angle E, \angle C = \angle F$ and

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

- The order of the letters shows which vertices correspond: $A \leftrightarrow D, B \leftrightarrow E, C \leftrightarrow F$.
- The common value of the ratio of corresponding sides is the **scale factor**.
- The ratio of the perimeters of two similar triangles equals the ratio of their corresponding sides.

Example 2.1 $\triangle ABC \sim \triangle DEF$ with $AB = 4$ cm, $BC = 5$ cm and $DE = 6$ cm. Find EF.

SOLUTION

$\frac{AB}{DE} = \frac{BC}{EF}$, so $\frac{4}{6} = \frac{5}{EF}$

$EF = \frac{5 \times 6}{4}$

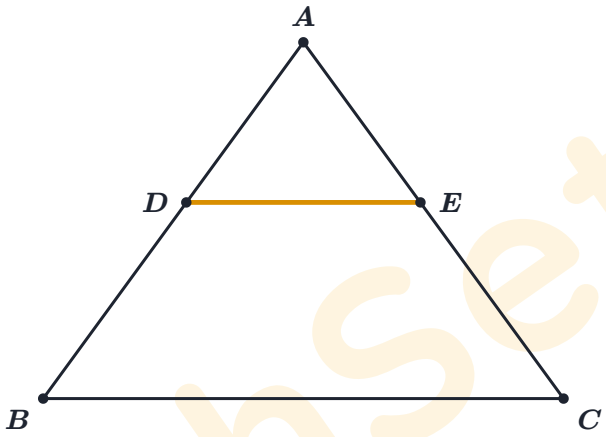
$EF = 7.5$ cm

3 Basic Proportionality Theorem (Thales theorem)

- **Theorem**: if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.
- In $\triangle ABC$, if $DE \parallel BC$ with D on AB and E on AC, then

$$\frac{AD}{DB} = \frac{AE}{EC}$$

- It also follows that $\frac{AD}{AB} = \frac{AE}{AC}$ and $\frac{DB}{AB} = \frac{EC}{AC}$.
- **Converse**: if a line divides any two sides of a triangle in the same ratio, the line is parallel to the third side.



DE is parallel to BC, so AD : DB = AE : EC

Example 3.1 In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm, find EC.

SOLUTION

By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$

$\frac{2}{3} = \frac{4}{EC}$

$EC = 6$ cm

Example 3.2 In $\triangle ABC$, $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, find x.

SOLUTION

$\frac{x}{x-2} = \frac{x+2}{x-1}$

$x(x-1) = (x+2)(x-2)$, so $x^2 - x = x^2 - 4$

$x = 4$

Example 3.3 D and E are points on sides AB and AC of $\triangle ABC$ with $AD = 4$ cm, $DB = 6$ cm, $AE = 6$ cm and $EC = 9$ cm. Is $DE \parallel BC$?

SOLUTION

$\frac{AD}{DB} = \frac{4}{6} = \frac{2}{3}$ and $\frac{AE}{EC} = \frac{6}{9} = \frac{2}{3}$

The two ratios are equal.

Yes. By the converse of the Basic Proportionality Theorem, $DE \parallel BC$.

4 Criteria for similarity of triangles

- **AAA** (or **AA**): if the corresponding angles of two triangles are equal, the triangles are similar. Two pairs of equal angles are enough, because the third pair must then be equal.
- **SSS**: if the corresponding sides of two triangles are in the same ratio, the triangles are similar.
- **SAS**: if one angle of a triangle equals one angle of another and the sides including these angles are in the same ratio, the triangles are similar.

Criterion	What to show
AA	Two pairs of corresponding angles are equal
SSS	$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$
SAS	$\angle A = \angle D$ and $\frac{AB}{DE} = \frac{AC}{DF}$

Example 4.1 The sides of one triangle are 3 cm, 4 cm and 5 cm, and the sides of another are 6 cm, 8 cm and 10 cm. Are the triangles similar?

SOLUTION

Compare the sides in order of size: $\frac{3}{6} = \frac{4}{8} = \frac{5}{10} = \frac{1}{2}$

Yes, the triangles are similar by the SSS criterion.

Example 4.2 In $\triangle ABC$ and $\triangle PQR$, $\angle A = \angle P = 50^\circ$, $AB = 6$ cm, $AC = 9$ cm, $PQ = 4$ cm and $PR = 6$ cm. Are the triangles similar?

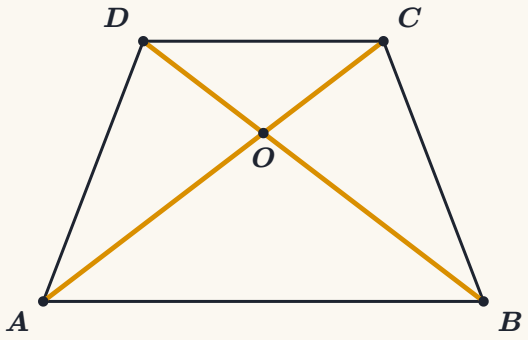
SOLUTION

$\frac{AB}{PQ} = \frac{6}{4} = \frac{3}{2}$ and $\frac{AC}{PR} = \frac{9}{6} = \frac{3}{2}$

The included angles are equal and the sides including them are in the same ratio.

Yes, $\triangle ABC \sim \triangle PQR$ by the SAS criterion.

Example 4.3 Diagonals AC and BD of a trapezium $ABCD$, with $AB \parallel DC$, intersect at O. Show that $\frac{OA}{OC} = \frac{OB}{OD}$.



Trapezium ABCD with AB parallel to DC

SOLUTION

In $\triangle OAB$ and $\triangle OCD$: $\angle OAB = \angle OCD$ (alternate angles, $AB \parallel DC$)

$\angle AOB = \angle COD$ (vertically opposite angles)

So $\triangle OAB \sim \triangle OCD$ by the AA criterion.

Hence $\frac{OA}{OC} = \frac{OB}{OD}$, since corresponding sides of similar triangles are in the same ratio.

5 Applications: heights and shadows

- At the same time of day, the sun's rays make the same angle with the ground, so an object and its shadow form similar right-angled triangles.

$$\frac{\text{height of object 1}}{\text{shadow of object 1}} = \frac{\text{height of object 2}}{\text{shadow of object 2}}$$

Example 5.1 A vertical pole 5 m high casts a shadow 3 m long. At the same time a tower casts a shadow 24 m long. Find the height of the tower.

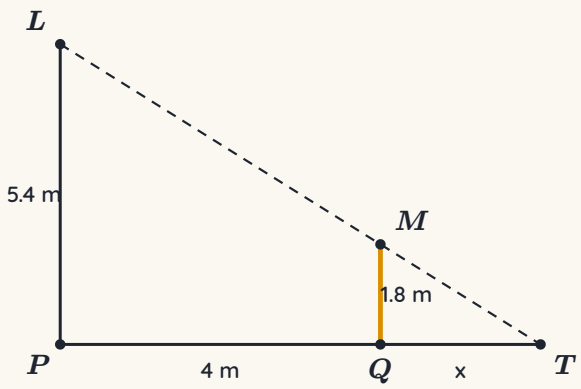
SOLUTION

The two triangles are similar by AA (a right angle each, and the same angle of the sun).

$\frac{h}{24} = \frac{5}{3}$

The tower is 40 m high.

Example 5.2 A man 1.8 m tall stands 4 m from the foot of a lamp-post 5.4 m high. Find the length of his shadow.



Lamp-post LP, man MQ and his shadow QT

SOLUTION

Let the shadow be x m. The triangle formed by the lamp-post and the triangle formed by the man share the angle at the tip of the shadow, and each has a right angle, so they are similar.

$\frac{5.4}{1.8} = \frac{4+x}{x}$, so $3x = 4 + x$

The shadow is 2 m long.

6 Writing a proof

- Name the two triangles and write the vertices in corresponding order.
- Give a reason for each equal angle or equal ratio: common angle, vertically opposite angles, alternate angles, corresponding angles, given.
- State the criterion used (AA, SSS or SAS), then write the ratio of the sides you need.

7 Common mistakes

- Writing the vertices in the wrong order, so that the wrong sides are matched.
- Using $\frac{AD}{DB} = \frac{DE}{BC}$. The ratio $\frac{DE}{BC}$ equals $\frac{AD}{AB}$, not $\frac{AD}{DB}$.
- Using SAS when the equal angle is not the angle between the two sides.
- Treating similar triangles as congruent.